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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 1

CLASS 8

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions × 2 marks = 80 marks
Section II	5 Star Questions × 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Operations · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling · Applied Mathematics · Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

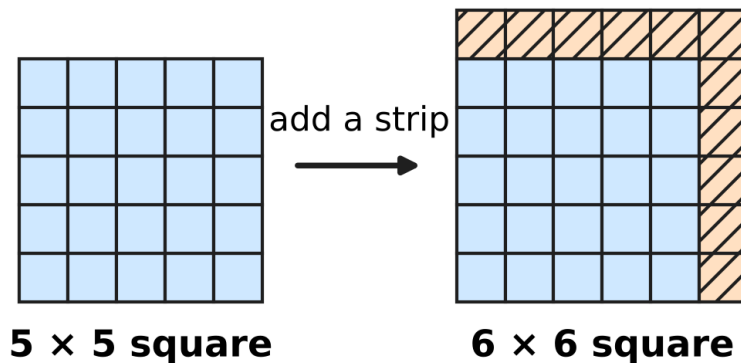
Parts follow the AMO syllabus chapters. Chapter 7 is not part of the Class 8 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)**

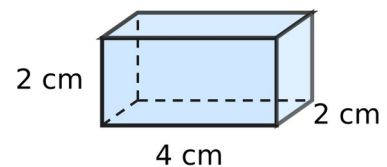
Q1. The picture shows a 5×5 square of tiles made into a 6×6 square by adding a strip of shaded tiles along two sides. In the same way, a 24×24 square of tiles is made into a 25×25 square. How many tiles are added?

- (A) 48 (B) 49 (C) 50 (D) 625

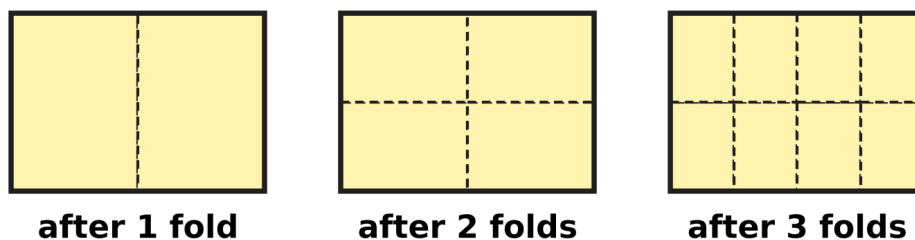
Q2. Which of these numbers lies between $\sqrt{3}$ and $\sqrt{5}$?

- (A) 1.7 (B) 1.9 (C) 2.3 (D) 4

Q3. Arun has many wooden blocks like the one in the picture, each $4 \text{ cm} \times 2 \text{ cm} \times 2 \text{ cm}$. He wants to put some of them together, without gaps or cutting, to make one solid cube. What is the least number of blocks he needs?



- (A) 2 (B) 4 (C) 16 (D) 32

Part 2 — Operations & Number Properties (Syllabus Chapter 2)

Q4. A sheet of paper is folded in half again and again. The picture shows the sheet opened out after 1, 2 and 3 folds; the dashed lines are the creases. The sheet is folded in half 6 times and opened out. What fraction of the whole sheet is each small part?

- (A) 2^6 (B) 6^{-2} (C) 12^{-1} (D) 2^{-6}

Q5. A fine grain of sand is 0.000075 m wide. How is this width written in standard form?

- (A) 7.5×10^{-5} m (B) 7.5×10^{-6} m
(C) 7.5×10^{-4} m (D) 7.5×10^5 m

Q6. How many of these six numbers are equal to $\frac{1}{16}$?

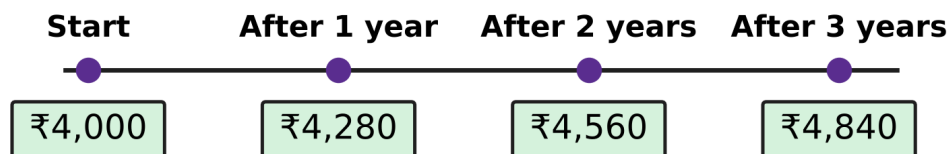
$$2^{-4}, 4^{-2}, (-4)^{-2}, 8^{-2}, \left(\frac{1}{4}\right)^{-2}, (-2)^{-4}$$

- (A) 2 (B) 3 (C) 4 (D) 5

Part 3 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

Q7. At a stationery shop, Asha sees the price tag in the picture. The price was cut twice, and each crossed-out price was the price before the next cut. The first cut was a 20% discount. What percentage was the second discount?

- (A) 32% (B) 85% (C) 12% (D) 15%



Q8. Navya puts some money in a savings scheme that pays simple interest. The picture shows the amount she has at the start and after each year. What is the rate of interest per year?

- (A) 7% (B) 14% (C) 21% (D) 28%

Q9. In the school vegetable garden, Ishan measures a bean plant. It is 50 cm tall. In the next week its height increases by 10%. In the week after that, its new height increases by 20%. How tall is the plant now?

- (A) 55 cm (B) 60 cm (C) 65 cm (D) 66 cm

Q10. At simple interest, a sum of money becomes $\frac{7}{4}$ of itself in 10 years. What is the rate of interest per year?

- (A) 7.5% (B) $13\frac{1}{3}\%$ (C) 17.5% (D) 75%

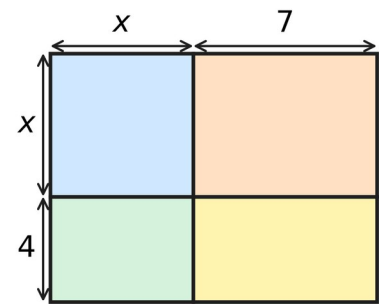
Q11. The number of books in a school library went up by 40% in one year. In the next year, old books were removed and the new number went down by 15%. The library now has 4,760 books. How many books did it have at the start?

- (A) 4,000 (B) 3,808 (C) 5,600 (D) 3,360

Part 4 — Algebra (Syllabus Chapter 4)

Q12. A rectangle is cut into four parts as shown in the picture. Its length is $(x + 7)$ and its width is $(x + 4)$. Which expression gives the area of the whole rectangle?

- (A) $x^2 + 28$ (B) $2x + 11$
 (C) $x^2 + 28x + 11$ (D) $x^2 + 11x + 28$



Q13. Three consecutive multiples of 8 add up to 264. What is the largest of the three?

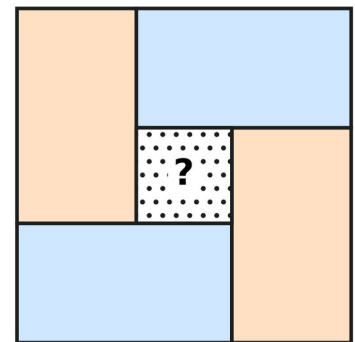
- (A) 80 (B) 88 (C) 96 (D) 104

Q14. Which of these is a factor of $x^2 - x - 30$?

- (A) $x - 6$ (B) $x + 6$ (C) $x - 10$ (D) $x - 5$

Q15. Four identical rectangular tiles are placed round a square hole to make a big square, as in the picture. The big square has area 196 cm^2 . Each tile has area 45 cm^2 . What is the side of the square hole?

- (A) 9 cm (B) 5 cm (C) 16 cm (D) 4 cm



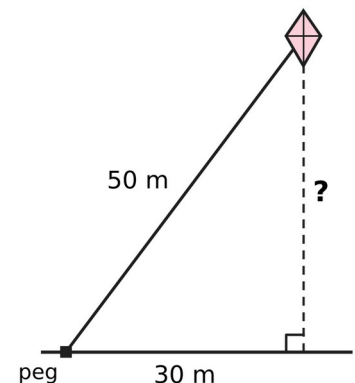
Q16. For a school bus outing, the fare for each child is ₹30 less than the fare for each adult. A group of 4 adults and 8 children pays ₹1,320 in all. What is the fare for one child?

- (A) ₹100 (B) ₹130 (C) ₹110 (D) ₹120

Part 5 — Geometry (Syllabus Chapter 5)

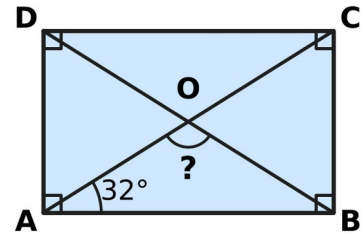
Q17. On a windy day, Uday ties his kite string to a peg in flat ground. The string is 50 m long and stretched straight. The kite is directly above a point 30 m from the peg, as in the picture. How high is the kite above the ground?

- (A) 25 m (B) 20 m (C) 80 m (D) 40 m



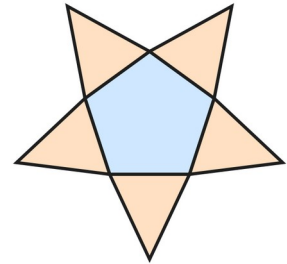
Q18. ABCD is a rectangle. Its diagonals AC and BD meet at O. $\angle OAB = 32^\circ$, as in the picture. What is $\angle AOB$?

- (A) 148° (B) 64° (C) 58° (D) 116°



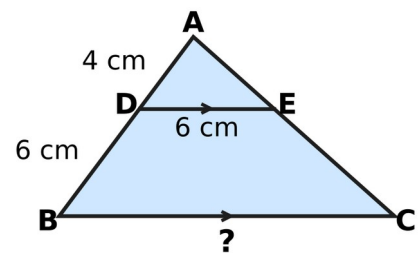
Q19. The picture shows a net made of one regular pentagon and five identical triangles. It is folded to make a closed solid. How many edges does the solid have?

- (A) 6 (B) 10 (C) 15 (D) 20



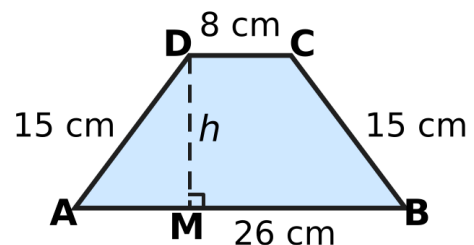
Q20. In $\triangle ABC$, D is on AB and E is on AC, with $DE \parallel BC$. $AD = 4$ cm, $DB = 6$ cm and $DE = 6$ cm, as in the picture. What is the length of BC?

- (A) 15 cm (B) 9 cm (C) 12 cm (D) 10 cm



Q21. ABCD is a trapezium with $DC \parallel AB$, as in the picture. $DC = 8$ cm, $AB = 26$ cm, and the slanting sides AD and BC are both 15 cm. DM is perpendicular to AB. What is the height $h = DM$?

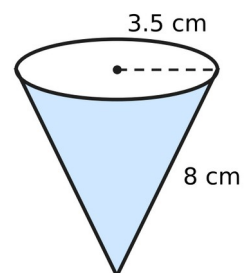
- (A) 9 cm (B) 6 cm (C) 12 cm (D) $\sqrt{306}$ cm



Part 6 — Mensuration (Syllabus Chapter 6)

Q22. Bela makes a cone-shaped paper cup, open at the top. Its top has radius 3.5 cm and its slant height is 8 cm, as in the picture. How much paper is needed for the cup? (Use $\pi = \frac{22}{7}$.)

- (A) 88 cm^2 (B) 126.5 cm^2
(C) 176 cm^2 (D) 308 cm^2

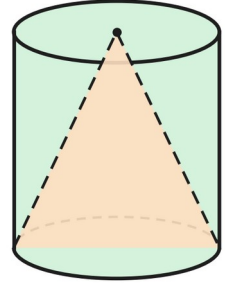


Q23. A classroom notice board is a rectangle 1.5 m long and 0.8 m wide. What is its area in square centimetres?

- (A) 120 cm^2 (B) $1,200 \text{ cm}^2$
 (C) $12,000 \text{ cm}^2$ (D) $120,000 \text{ cm}^2$

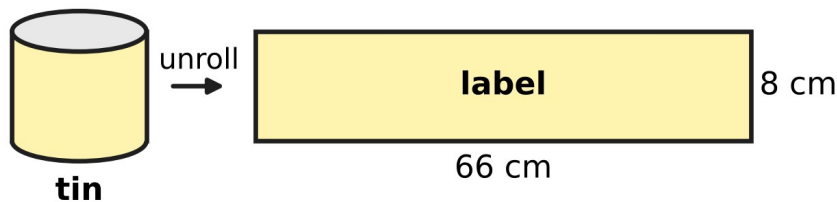
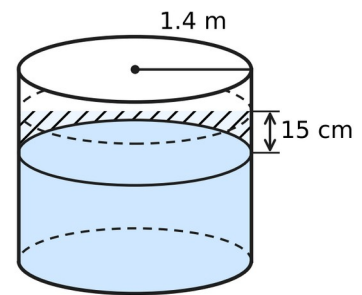
Q24. A solid wooden cylinder has volume 840 cm^3 . A cone with the same base and the same height as the cylinder is carved out of it, as shown by the dashed lines in the picture. What is the volume of the wood that is left?

- (A) 280 cm^3 (B) 420 cm^3 (C) 560 cm^3 (D) 630 cm^3



Q25. A rainwater tank is a cylinder with radius 1.4 m. Some water is taken out and the water level falls by 15 cm, as in the picture. How many litres of water were taken out? (Use $\pi = \frac{22}{7}$; $1 \text{ m}^3 = 1,000 \text{ litres}$.)

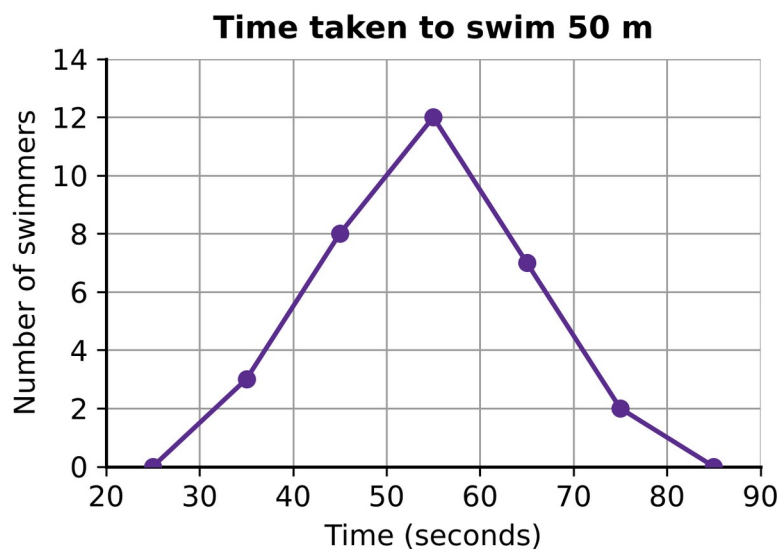
- (A) 92.4 litres (B) 3,696 litres
 (C) 9,240 litres (D) 924 litres



Q26. A paper label covers the whole curved surface of a closed cylindrical tin, with no overlap. When the label is taken off and laid flat, it is a rectangle 66 cm long and 8 cm wide, as in the picture. What is the volume of the tin? (Use $\pi = \frac{22}{7}$.)

- (A) $2,772 \text{ cm}^3$ (B) 264 cm^3
 (C) 528 cm^3 (D) $11,088 \text{ cm}^3$

Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)



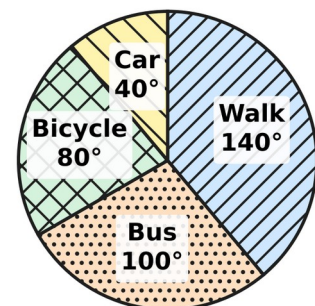
Q27. The swimmers at a swimming pool were timed over 50 m. Their times were grouped in classes 30–40, 40–50, 50–60, 60–70 and 70–80 seconds, where 30–40 means 30 or more but less than 40. The frequency polygon in the picture shows the number of swimmers in each class, plotted at the middle of the class. How many swimmers took 60 seconds or more?

- (A) 7 (B) 9 (C) 12 (D) 21

Q28. The pie chart shows how the 540 students of a school come to school. The angle of each sector is marked. How many more students walk to school than come by bicycle?

- (A) 60 (B) 90 (C) 120 (D) 210

How students come to school



Total: 540 students

Goals scored by a school team in 25 matches

Goals in a match	0	1	2	3	4	5
Number of matches	4	7	6	4	3	1

Q29. The table shows how many goals a school team scored in each of its 25 matches. What is the median number of goals scored in a match?

- (A) 1 (B) 2 (C) 2.5 (D) 4

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Q30. Kartik buys a used bicycle for ₹2,400 and spends ₹600 on repairs. He then sells it for ₹3,600. What is his profit percent?

- (A) 25% (B) $16\frac{2}{3}\%$ (C) 20% (D) 50%

Q31. A sum of money earns ₹540 as simple interest in 3 years. How much simple interest will the same sum earn in 5 years at the same rate?

- (A) ₹720 (B) ₹900 (C) ₹1,080 (D) ₹2,700

Q32. Leela sells a glass jar full of marbles for ₹780 and makes a profit of 20%. What did the jar of marbles cost her?

- (A) ₹624 (B) ₹936 (C) ₹130 (D) ₹650

Q33. Farhan buys 50 kg of apples from an orchard at ₹80 per kg. Of these, 10% rot and are thrown away. He sells the rest at ₹100 per kg. What is his profit percent?

- (A) 10% (B) 12.5% (C) 15% (D) 25%

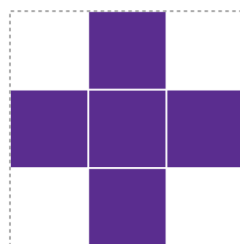
Q34. For her stamp collection club, Swati buys 50 stamps at ₹12 each and 50 stamps at ₹18 each. She sells the cheaper stamps at a profit of 25% and the costlier stamps at a loss of 10%. What is her overall profit or loss percent?

- (A) 4% profit (B) 4% loss
(C) 7.5% profit (D) 15% profit

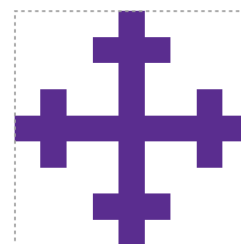
Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)



Step 0



Step 1



Step 2

Q35. The picture shows the first steps of a fractal. At each step, every filled square is cut into 9 equal small squares, and only the middle one and the four touching its sides are kept. How many filled squares will there be at Step 3?

- (A) 25 (B) 45 (C) 125 (D) 625

Q36. In the number 74,3d6, the letter d stands for one digit. Which of these values of d makes the number divisible by 8?

- (A) 1 (B) 5 (C) 7 (D) 9

Q37. A 4-digit palindrome reads the same from left to right and from right to left, like 2,662. Which number divides EVERY 4-digit palindrome?

- (A) 7 (B) 11 (C) 13 (D) 101

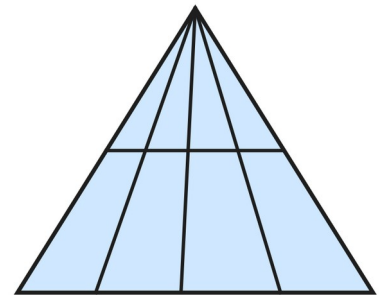
Q38. In the multiplication shown in the picture, each letter stands for a digit. Different letters stand for different digits, and no number starts with 0. What is the two-digit number AB?

- (A) 95 (B) 57 (C) 75 (D) 35

$$\begin{array}{r} \text{A B} \\ \times \quad \text{A} \\ \hline \text{B C B} \end{array}$$

Q39. In the picture, three lines are drawn from the top corner of a big triangle to its base, and one more line is drawn across it from side to side. How many triangles of all sizes are there in the picture?

- (A) 8 (B) 10 (C) 14 (D) 20



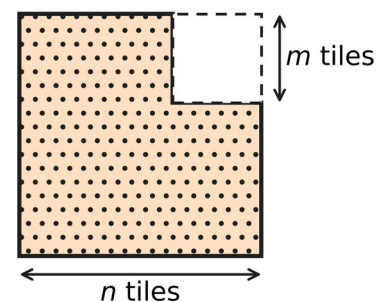
Q40. Advait's farm has some cows and some hens, at least one of each. Together they have 35 heads. Which of these could be the total number of their legs?

- (A) 64 (B) 67 (C) 98 (D) 142

SECTION II — Star Questions 5 questions × 4 marks each

Q41. A square floor is made of $n \times n$ square tiles. A smaller square of $m \times m$ tiles is removed from one corner, as in the picture, where m is at least 1. Exactly 96 tiles are left. How many different pairs of whole numbers (n, m) are possible? [4 marks]

- (A) 4 (B) 6 (C) 8 (D) 12

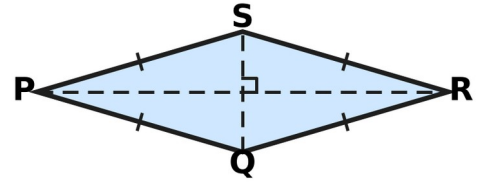


Q42. A solid has flat faces and no holes, so $F + V - E = 2$ for its faces F , vertices V and edges E . Every face is a triangle or a square. Every edge joins two vertices and is shared by exactly two faces. The solid has 24 vertices, and exactly 4 edges meet at every vertex. How many of its faces are squares? [4 marks]

- (A) 8 (B) 13 (C) 18 (D) 24

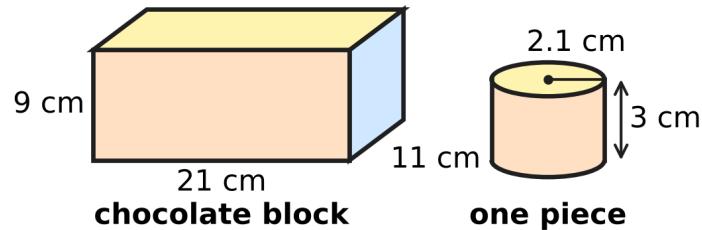
Q43. PQRS is a rhombus with perimeter 100 cm. Its diagonals PR and QS, shown in the picture, differ in length by 34 cm. What is the area of the rhombus? [4 marks]

- (A) 168 cm^2 (B) 336 cm^2 (C) 625 cm^2 (D) 672 cm^2



Q44. Samar borrows ₹20,000 at 9% per year simple interest and uses it all to buy school bags. He marks the bags 25% above his cost and sells all of them at a discount of 8% on the marked price. After 8 months he repays the loan with its interest. What is his profit as a percentage of ₹20,000? [4 marks]

- (A) 6% (B) 9% (C) 11% (D) 15%



Q45. At a bakery, Reena melts a cuboid block of chocolate and pours all of it into cylindrical moulds. Each mould makes one piece, and no chocolate is lost. The sizes are shown in the picture. The block costs ₹1,600 and each piece is wrapped in a paper that costs ₹4. She wants a profit of 25% on her total cost. At what price should she sell each wrapped piece? (Use $\pi = \frac{22}{7}$.) [4 marks]

- (A) ₹36 (B) ₹40 (C) ₹44 (D) ₹45

Answer Sheet AMO Level 1 • Class 8 • Practice Paper 1**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 · Class 8 · Practice Paper 1**Quick answers**

1 B	6 C	11 A	16 A	21 C	26 A	31 B	36 C	41 A
2 B	7 D	12 D	17 D	22 A	27 B	32 D	37 B	42 C
3 B	8 A	13 C	18 D	23 C	28 B	33 B	38 C	43 B
4 D	9 D	14 A	19 B	24 C	29 B	34 A	39 D	44 B
5 A	10 A	15 D	20 A	25 D	30 C	35 C	40 C	45 D

Solutions

Q	Part · level	Answer	Why?
1	Part 1 · Ch1 · Easy	(B) 49	Tiles added = $25^2 - 24^2 = 625 - 576 = 49$ (that is, $24 + 24 + 1$ for the corner).
2	Part 1 · Ch1 · Easy	(B) 1.9	Square each number: $1.9^2 = 3.61$ lies between 3 and 5, so $\sqrt{3} < 1.9 < \sqrt{5}$. But $1.7^2 = 2.89$, $2.3^2 = 5.29$ and $4^2 = 16$ do not.
3	Part 1 · Ch1 · Medium	(B) 4	Volume of a block = 16 cm^3 . The smallest cube whose volume is a multiple of 16 and whose edge can hold a 4 cm length is $4 \text{ cm} \times 4 \text{ cm} \times 4 \text{ cm} = 64 \text{ cm}^3$, and $64 \div 16 = 4$ blocks.
4	Part 2 · Ch2 · Easy	(D) 2^{-6}	Each fold doubles the number of parts: 2, 4, 8, ... so 6 folds give $2^6 = 64$ parts. One part is $1/2^6 = 2^{-6}$ of the sheet.
5	Part 2 · Ch2 · Easy	(A) $7.5 \times 10^{-5} \text{ m}$	Move the decimal point 5 places to the right to get 7.5, so $0.000075 = 7.5 \times 10^{-5} \text{ m}$.
6	Part 2 · Ch2 · Medium	(C) 4	Here $2^{-4} = 1/16$, $4^{-2} = 1/16$, $(-4)^{-2} = 1/16$ and $(-2)^{-4} = 1/16$. But $8^{-2} = 1/64$ and $(1/4)^{-2} = 16$. So 4 of them equal $1/16$.
7	Part 3 · Ch3 · Easy	(D) 15%	The second cut is $640 - 544 = ₹96$ on a price of ₹640. As a percentage, $96/640 \times 100 = 15\%$.
8	Part 3 · Ch3 · Easy	(A) 7%	The amount grows by ₹280 every year, so the interest is ₹280 per year on ₹4,000. Rate = $280/4,000 \times 100 = 7\%$ per year.
9	Part 3 · Ch3 · Easy	(D) 66 cm	After week 1: $50 \times 1.1 = 55 \text{ cm}$. After week 2: $55 \times 1.2 = 66 \text{ cm}$.
10	Part 3 · Ch3 · Medium	(A) 7.5%	Interest = $7/4 - 1 = 3/4$ of the sum in 10 years, so $3/40$ of the sum each year. Rate = $3/40 \times 100 = 7.5\%$ per year.
11	Part 3 · Ch3 · Medium	(A) 4,000	A rise of 40% then a fall of 15% multiply the number by $1.4 \times 0.85 = 1.19$. So the start = $4,760 \div 1.19 = 4,000$ books.
12	Part 4 · Ch4 · Easy	(D) $x^2 + 11x + 28$	The four parts are x^2 , $7x$, $4x$ and $4 \times 7 = 28$. Together: $x^2 + 11x + 28$.
13	Part 4 · Ch4 · Easy	(C) 96	Let them be $8k - 8$, $8k$ and $8k + 8$. Their sum is $24k = 264$, so $k = 11$. The numbers are 80, 88 and 96; the largest is 96.
14	Part 4 · Ch4 · Easy	(A) $x - 6$	Two numbers with product -30 and sum -1 are -6 and 5 . So $x^2 - x - 30 = (x - 6)(x + 5)$, and $x - 6$ is a factor.
15	Part 4 · Ch4 · Medium	(D) 4 cm	If a tile is a by b , the big square has side $a + b$ and the hole has side $a - b$. $(a - b)^2 = (a + b)^2 - 4ab = 196 - 4 \times 45 = 16$, so the side is 4 cm.
16	Part 4 · Ch4 · Medium	(A) ₹100	Let a child pay ₹ c , so an adult pays ₹ $(c + 30)$. Then $4(c + 30) + 8c = 1,320$, so $12c = 1,200$ and $c = 100$.
17	Part 5 · Ch5 · Easy	(D) 40 m	The string is the hypotenuse: $\text{height}^2 = 50^2 - 30^2 = 2,500 - 900 = 1,600$, so the height is 40 m.

Q	Part · level	Answer	Why?
18	Part 5 · Ch5 · Easy	(D) 116°	The diagonals of a rectangle are equal and bisect each other, so $OA = OB$ and $\angle OBA = \angle OAB = 32^\circ$. Then $\angle AOB = 180^\circ - 32^\circ - 32^\circ = 116^\circ$.
19	Part 5 · Ch5 · Easy	(B) 10	The net folds into a pentagonal pyramid: 5 edges round the base and 5 edges up to the top, so 10 edges. (Check: $F + V - E = 6 + 6 - 10 = 2$.)
20	Part 5 · Ch5 · Medium	(A) 15 cm	$DE \parallel BC$ gives equal corresponding angles, so $\triangle ADE \sim \triangle ABC$. $AB = 4 + 6 = 10$ cm, so $BC/DE = AB/AD = 10/4$ and $BC = 6 \times 10/4 = 15$ cm.
21	Part 5 · Ch5 · Medium	(C) 12 cm	The two slanting sides are equal, so $AM = (26 - 8) \div 2 = 9$ cm. In right $\triangle AMD$, $h^2 = 15^2 - 9^2 = 225 - 81 = 144$, so $h = 12$ cm.
22	Part 6 · Ch6 · Easy	(A) 88 cm ²	Paper = curved surface = $\pi rl = 22/7 \times 3.5 \times 8 = 88$ cm ² .
23	Part 6 · Ch6 · Easy	(C) 12,000 cm ²	Area = $1.5 \times 0.8 = 1.2$ m ² . Since 1 m ² = 100 cm × 100 cm = 10,000 cm ² , the area is 12,000 cm ² .
24	Part 6 · Ch6 · Easy	(C) 560 cm ³	A cone with the same base and height has 1/3 of the cylinder's volume: $840 \div 3 = 280$ cm ³ . Wood left = $840 - 280 = 560$ cm ³ .
25	Part 6 · Ch6 · Medium	(D) 924 litres	Water taken out = $\pi r^2 h = 22/7 \times 1.4 \times 1.4 \times 0.15 = 0.924$ m ³ = 924 litres.
26	Part 6 · Ch6 · Medium	(A) 2,772 cm ³	The 66 cm side goes round the tin, so $2\pi r = 66$ and $r = 10.5$ cm. The height is 8 cm. Volume = $22/7 \times 10.5 \times 10.5 \times 8 = 2,772$ cm ³ .
27	Part 7 · Ch8 · Easy	(B) 9	Taking 60 seconds or more means the classes 60–70 and 70–80. Their points (at 65 and 75) are at 7 and 2, so $7 + 2 = 9$ swimmers.
28	Part 7 · Ch8 · Easy	(B) 90	The difference in angle is $140^\circ - 80^\circ = 60^\circ$. That is $60/360$ of 540 students = 90 students.
29	Part 7 · Ch8 · Medium	(B) 2	With 25 matches, the median is the 13th value in order. The first 4 values are 0 and the next 7 are 1 (11 values so far); the 12th to 17th values are 2. So the median is 2.
30	Part 8 · Ch9 · Easy	(C) 20%	Cost price = $2,400 + 600 = ₹3,000$. Profit = $3,600 - 3,000 = ₹600$. Profit percent = $600/3,000 \times 100 = 20\%$.
31	Part 8 · Ch9 · Easy	(B) ₹900	Simple interest is the same every year: $₹540 \div 3 = ₹180$ per year. For 5 years: $180 \times 5 = ₹900$.
32	Part 8 · Ch9 · Easy	(D) ₹650	Selling price = 120% of the cost price. So the cost price = $780 \times 100/120 = ₹650$.
33	Part 8 · Ch9 · Medium	(B) 12.5%	Cost = $50 \times 80 = ₹4,000$. He sells 45 kg for $45 \times 100 = ₹4,500$. Profit = ₹500, which is $500/4,000 \times 100 = 12.5\%$.
34	Part 8 · Ch9 · Medium	(A) 4% profit	Cost = $600 + 900 = ₹1,500$. Selling price = $600 \times 1.25 + 900 \times 0.9 = 750 + 810 = ₹1,560$. Profit = ₹60, which is $60/1,500 \times 100 = 4\%$.
35	Part 9 · Ch10 · Easy	(C) 125	Each filled square becomes 5 filled squares, so the number is multiplied by 5 at every step: 1, 5, 25, 125. At Step 3 there are $5^3 = 125$ squares.
36	Part 9 · Ch10 · Easy	(C) 7	A number is divisible by 8 when the number made by its last three digits is. Of the options, only $376 = 8 \times 47$ works; 316, 356 and 396 are not multiples of 8. So $d = 7$.
37	Part 9 · Ch10 · Easy	(B) 11	The example $2,662 = 2 \times 11 \times 11 \times 11$ is not a multiple of 7, 13 or 101, so only 11 can work. It always does: $abba = 1,001a + 110b = 11 \times (91a + 10b)$.

Q	Part · level	Answer	Why?
38	Part 9 · Ch10 · Medium	(C) 75	$B \times A$ must end in B. $A = 1$ gives a two-digit product, so either $A = 6$ with B even, or $B = 5$ with A odd. $A = 6$ gives $62 \times 6 = 372$, $64 \times 6 = 384$, $68 \times 6 = 408$: none fits. $B = 5$ gives $35 \times 3 = 105$, $75 \times 7 = 525$, $95 \times 9 = 855$; only 525 has the form BCB, so $AB = 75$.
39	Part 9 · Ch10 · Medium	(D) 20	Every triangle has the top corner and two of the 5 lines from it, with its base on the cross line or on the bottom line. Choosing 2 of the 5 lines gives 10 ways, and there are 2 bases: $2 \times 10 = 20$.
40	Part 9 · Ch10 · Medium	(C) 98	Each animal has 2 or 4 legs, so the total is even. The fewest legs come from 1 cow and 34 hens: $4 + 68 = 72$. The most come from 34 cows and 1 hen: $136 + 2 = 138$. The total 98 works: 14 cows and 21 hens have $56 + 42 = 98$ legs.
41	Section II · Ch4 + Ch10 · Hard	(A) 4	$n^2 - m^2 = (n - m)(n + m) = 96$. The two factors differ by 2m, so both are even: 2×48 , 4×24 , 6×16 , 8×12 . These give $(n, m) = (25, 23)$, $(14, 10)$, $(11, 5)$, $(10, 2)$: 4 pairs.
42	Section II · Ch5 + Ch10 · Hard	(C) 18	Counting edge ends at the vertices, $E = 24 \times 4 \div 2 = 48$, so $F = 2 - 24 + 48 = 26$. Counting edges round the faces, $3 \times (\text{triangles}) + 4 \times (\text{squares}) = 2 \times 48 = 96$; with 26 faces this gives 8 triangles and 18 squares. (Such a solid really exists: the rhombicuboctahedron has exactly these faces.)
43	Section II · Ch4 + Ch5 · Hard	(B) 336 cm^2	Side = 25 cm. The diagonals bisect at right angles; let the half-diagonals be a and b, so $a^2 + b^2 = 625$ and $a - b = 17$. Then $2ab = 625 - 17^2 = 336$. The area is $4 \times \frac{1}{2}ab = 2ab = 336 \text{ cm}^2$.
44	Section II · Ch3 + Ch9 · Hard	(B) 9%	Selling price = $20,000 \times 1.25 \times 0.92 = ₹23,000$. Interest = $20,000 \times 9/100 \times 8/12 = ₹1,200$. Profit = $23,000 - 20,000 - 1,200 = ₹1,800$, which is 9% of ₹20,000.
45	Section II · Ch6 + Ch9 · Hard	(D) ₹45	Block = $21 \times 11 \times 9 = 2,079 \text{ cm}^3$. One piece = $22/7 \times 2.1 \times 2.1 \times 3 = 41.58 \text{ cm}^3$, so $2,079 \div 41.58 = 50$ pieces. Total cost = $1,600 + 50 \times 4 = ₹1,800$. With 25% profit she needs ₹2,250, that is ₹45 per piece.